eedi-mining-misconceptions-in-mathematics

10st Place Solution Summary

Rank	10
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Topic ID	551722
Version	Original

Source: https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/551722
Retrieved with Kaggle CLI on 2026-10-21.

Thanks to the organizers and Kaggle, this is my second solo gold medal , which is enough to make me a Kaggle Grandmaster. Below is a brief summary of my solution.

External data

7k external data using GPT-4O.

Retrievers

Model: Qwen/Qwen2.5-14B-Instruct,Qwen/Qwen2.5-Coder-14B-Instruct
Train detail:
firstly train(lora) with 7k additional data and then load the lora weight,lastly fintune the train data.
CV: train with 5 folds and then merge using lora.
I integrated a total of three recall models, including the 14b 5-folds, 14b-coder 5-folds, and the full 14b 5-folds.
The training code needs to be slightly modified in some Model classes from my previously open-sourced KDD code.

Rerankers

Model:Qwen32b AWQ,training with lora
Generated approximately For ranking, I used a multiple-choice approach.
The input consists of questions and answers, along with the top 25 options, and the output is the probabilities of these 25 options.
During training, I only optimized the probabilities of these 25 characters, using the cross-entropy loss function.
In the end, my leaderboard score was achieved using only the 5-folds (7k additional data + training data) of LoRA fusion (LoRA fusion remains effective in any scenario).

Try but not worked

Distillation：I tried to distill the content of 72b, but since I didn't see any improvement offline, I didn't submit it. I might have been wrong, after all, the offline validation data is relatively limited.

The first-place solution made me realize that there is still much room for optimization in my current solution, but I am also quite satisfied～